

Education

Computer Science and Technology **Sun Yat-sen University** **Aug. 2023 - Jun. 2027(Expected)**

- **Coursework:** Artificial Intelligence; Machine Learning; Optimization Theory; Data Structures and Algorithms; Mathematical analysis I & II;
- **GPA:** 3.8/4.0

Internship Experience

Tencent Holdings Limited **Backend Development Intern** **Mar. 2026 – Jun. 2026**

- Investigated performance optimization for an internal TensorFlow-to-PyTorch recommendation model training framework, leveraging cutting-edge deep learning compiler and distributed primitives.
- Profiled and evaluated the experimental thread-safe (GIL-protected) `torch.compile` by managing environment alignment and updating to the latest PyTorch versions, identifying specific open-source framework compiler bugs that limited runtime speedup.
- Diagnosed distributed training bottlenecks via precise PyTorch Profiler traces; designed an adaptive gradient accumulation mechanism across multi-step GPU executions to minimize collective communication overhead.
- Discovered that embedding layer operations, rather than pure `all_reduce` operations, dominated communication latency, providing critical empirical insights for non-universal communication bottlenecks.

Sponge Lab at HKUST **Research Intern** **Feb. 2026 – Present**

- Accelerated Vision-Language-Action (VLA) models and embodied AI control policies through model compression, non-structured pruning, and low-precision quantization.
- Developed custom visualization pipelines for an ongoing ECCV submission that introduces a token pruning and biased attention mechanism for VLA model inference acceleration.
- Migrated the proposed ECCV token pruning framework onto the World Action Model (WAM) setup using the autoregressive DreamZero architecture to explore its operational transferability.
- Evaluated KIVI-style KV cache quantization on the `cosmos-policy` environment following its verified gains on DreamZero; systematically proved its architectural incompatibility due to the inherent absence of KV cache structures in non-autoregressive control policies.
- Pioneered the implementation 2:4 structured sparsity on Lingbot-Map to squeeze its hardware-level tensor core compute throughput.

Research & Project Experience

Optimization of code_saturne **ISC25' SCC Competition** **Mar. 2025 – May 2025**

- Addressed communication bottlenecks via MPI implementation comparisons and disk I/O minimization strategies.
- Achieved a significant 2.7 times performance improvement for the `LES_AIR_QUALITY` case on Bridges 2.
- Fixed GPU runtime errors; scalability requires architectural redesign.

Optimization of llama.cpp **ISC26' SCC Competition** **Mar. 2026 – June 2026**

- Implemented Data Parallelism (DP) across multi-GPU/multi-core CPU servers, achieving near-linear scalability 10x speedup by balancing computation loads.
- Developed custom Mixture-of-Experts (MoE) kernel fusion strategies to reduce memory bandwidth bottlenecks and minimize kernel launch overheads.
- Optimized Tensor Parallelism (TP) communication efficiency on multi-node architectures through systematic parameter tuning and communication-computation overlapping.

vLLM-Ascend Sampler

2026 Huawei ICT (Initial Round)

Nov. 2025 – Jan. 2026

- Identified performance bottlenecks in Top-K / Top-P sampling during LLM inference on Ascend NPUs.
- Implemented Triton-based fused sampling kernels to reduce kernel launches and memory access overhead.
- Achieved up to 1.88% latency reduction compared to the native Ascend 910 implementation.

Estimate Relative Camera Pose

Sep. 2025 – Feb. 2026

- Designed a minimal solver for relative pose estimation using point correspondences and surface normals.
- Evaluated robustness and efficiency with several benchmarks.

Awards

- ISC 2025 Student Cluster Competition(ISC25' SCC) 1st Place Online Winner
- ICT 2026 Challenge Track Third Prize
- 2023-2025 Sun Yat-sen University Scholarship

Languages and Technologies

- **Languages:** English (CET-6: 544); Mandarin (Native); Cantonese (Native)
- **Programming:** Python (familiar); C/C++ (familiar); OpenMP (familiar); MPI (familiar); CUDA (learning)
- **Systems & Tools:** Linux; GDB; CMake; Performance profiling and benchmarking; Docker